

Customer Segmentation Using K-Means Clustering

Introduction

Customer segmentation helps businesses understand different groups of customers based on their purchasing behavior. In this project, the Mall Customer Dataset was analyzed using the K-Means Clustering algorithm to identify customer segments based on income and spending patterns.

Objective

The objective of this project was to:

- Group customers with similar behavior.
- Identify different customer segments.
- Visualize customer clusters.
- Generate insights that can support marketing and business decisions.

Dataset

The Mall Customer Dataset contains information about mall visitors.

Features Used

- Annual Income (k\$)
- Spending Score (1–100)

These features were selected because they directly reflect customer purchasing behavior.

Methodology

1. Data Loading and Exploration

The dataset was loaded using Pandas and explored to understand its structure, columns, and data types.

2. Data Preprocessing

The selected features were scaled using StandardScaler to prepare the data for clustering.

3. Finding the Optimal Number of Clusters

The Elbow Method was used to determine the appropriate number of customer groups.

4. K-Means Clustering

The K-Means algorithm was applied to divide customers into different clusters based on income and spending score.

5. Visualization

Scatter plots were created to visualize and compare the customer segments.

Results and Insights

The analysis identified several customer groups:

- High Income and High Spending customers (premium customers)
- High Income and Low Spending customers (potential growth opportunities)
- Low Income and High Spending customers (active buyers)
- Low Income and Low Spending customers (budget-conscious customers)

These customer segments can help businesses create targeted marketing strategies and improve customer engagement.

Tools and Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Conclusion

This project successfully applied K-Means Clustering to segment customers based on their spending behavior. The generated customer groups provide useful insights that can help businesses better understand their customers and make data-driven decisions.